Ex. No. 1.2 – Design and Draft the Given 2D Sketches in Modelling Software

AIM

To sketch, dimension, model and draft the given component using Autodesk Fusion 360.

PROCEDURE

1. Create a new design and set units. Open Autodesk Fusion 360, create a new design, save it as “Ex 1.2”, and confirm that the document units are millimetres (mm).

2. Start the main sketch. Select Create Sketch and choose a suitable origin plane. Use the Line, Circle and Arc tools to draw the central circular region and two side portions. Position the central feature near the origin and build the remaining geometry around it.

3. Draw the central circular feature. Create concentric circles for the central opening and surrounding ring. Apply the diameters shown in the sketch, including Ø44 mm, Ø40 mm and Ø32 mm where indicated. Draw the radial slots using lines and arcs, then trim extra portions to form the segmented central opening.

4. Create the side circular features. Draw the left and right circular lugs and their concentric holes. Apply Ø30 mm to the side circular features where shown. Position the lug centres using the locating dimensions in the sketch, including 50 mm, 60 mm, 70 mm and 20 mm.

5. Form the outer lobed profile. Use tangent arcs and lines to connect the upper and lower lobes to the side lugs and central body. Apply the radii marked R32 mm, R40 mm, R20 mm and R21 mm to the corresponding curved sections. Use the 11.07 mm dimension for the indicated slanted segment or offset.

6. Add positional dimensions and constraints. Use Sketch Dimension (D) to apply the 26.66 mm and 15.59 mm measurements and other locating dimensions shown. Add Coincident, Concentric, Tangent, Horizontal/Vertical and Equal constraints where appropriate so connected edges meet cleanly.

7. Check and finish the 2D sketch. Inspect the profile for gaps, overlapping lines and unwanted arc sections. Confirm that the central slots, side holes and outer lobes match the reference. Select Finish Sketch when the intended outer profile is closed.

8. Extrude the profile to thickness. Select the closed profile and use Extrude. Set the thickness to 20 mm as shown in the side view. Keep the central opening, radial slots and side holes open; if required, cut them through the body using a sketch and Extrude > Cut.

9. Create the drawing views and add dimensions. Create a drawing from the model and place front, side and isometric views. Add the dimensions visible on the supplied drawing, including 162 mm, 132.65 mm, 19.3 mm and 12.05 mm in the upper side view, and 178.14 mm, 92.07 mm and 23.09 mm in the lower side view. Retain the 20 mm thickness dimension.

10. Inspect and save. Compare the model and drawing with the references. Check that the central segmented opening, both side lugs, curved outer lobes and through-holes are present. Save the Fusion 360 design and technical drawing.

SKETCHES

Initial 2D sketch in Fusion 360:

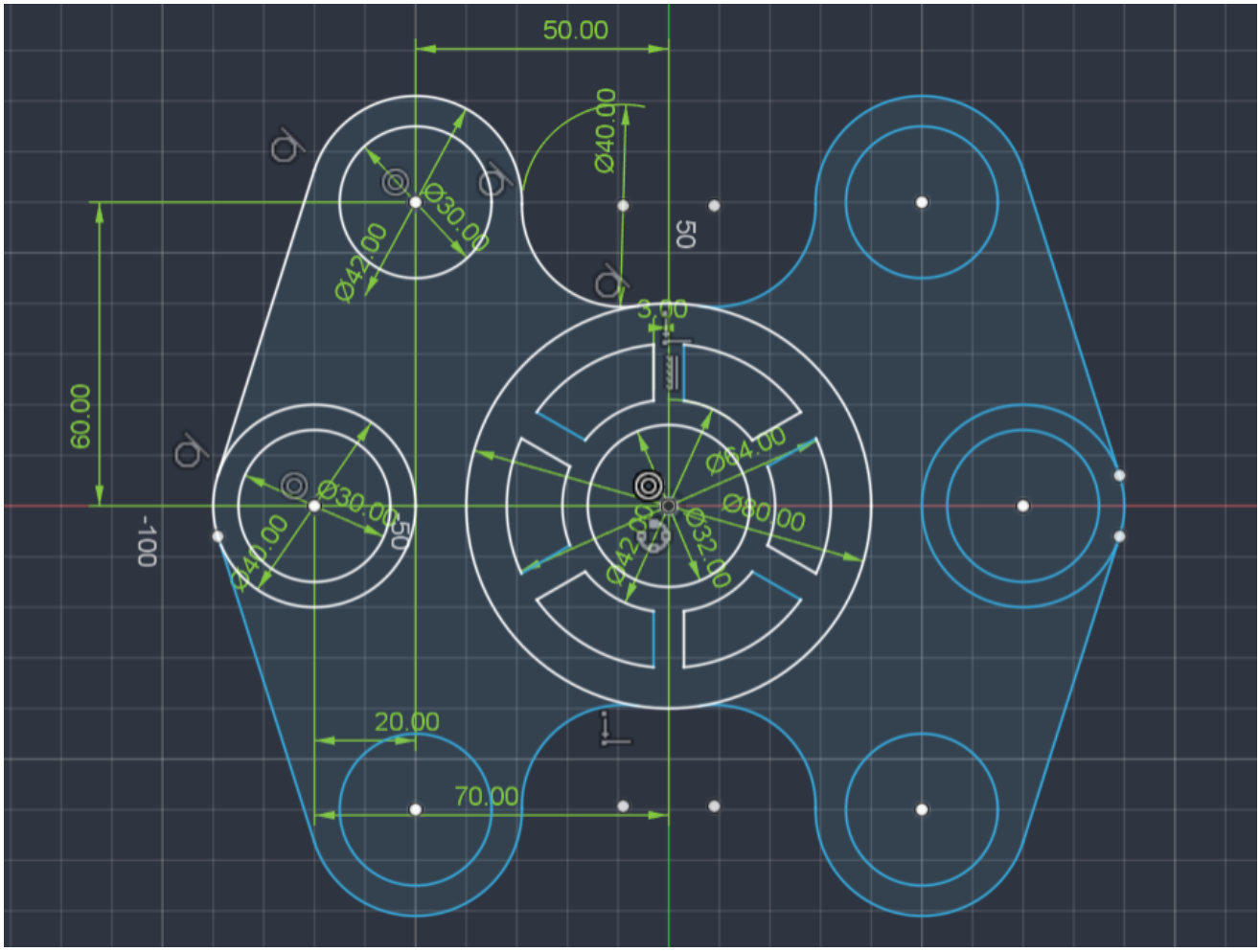


Figure 1: Dimensioned 2D sketch

3D model view:

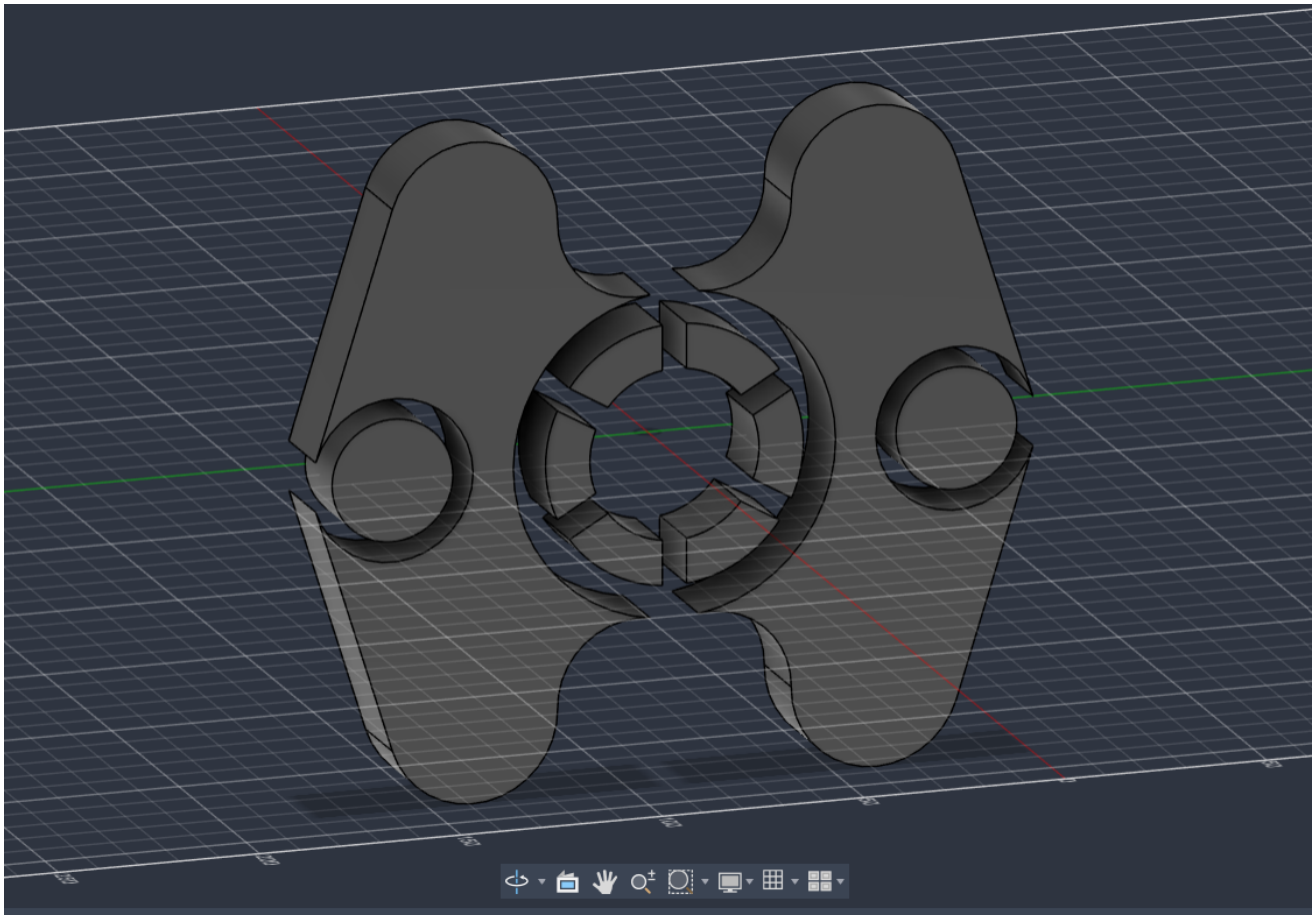


Figure 2: 3D model created in Fusion 360

OUTPUT

The component was modelled in Autodesk Fusion 360 and the technical drawing was prepared with the required views and dimensions.

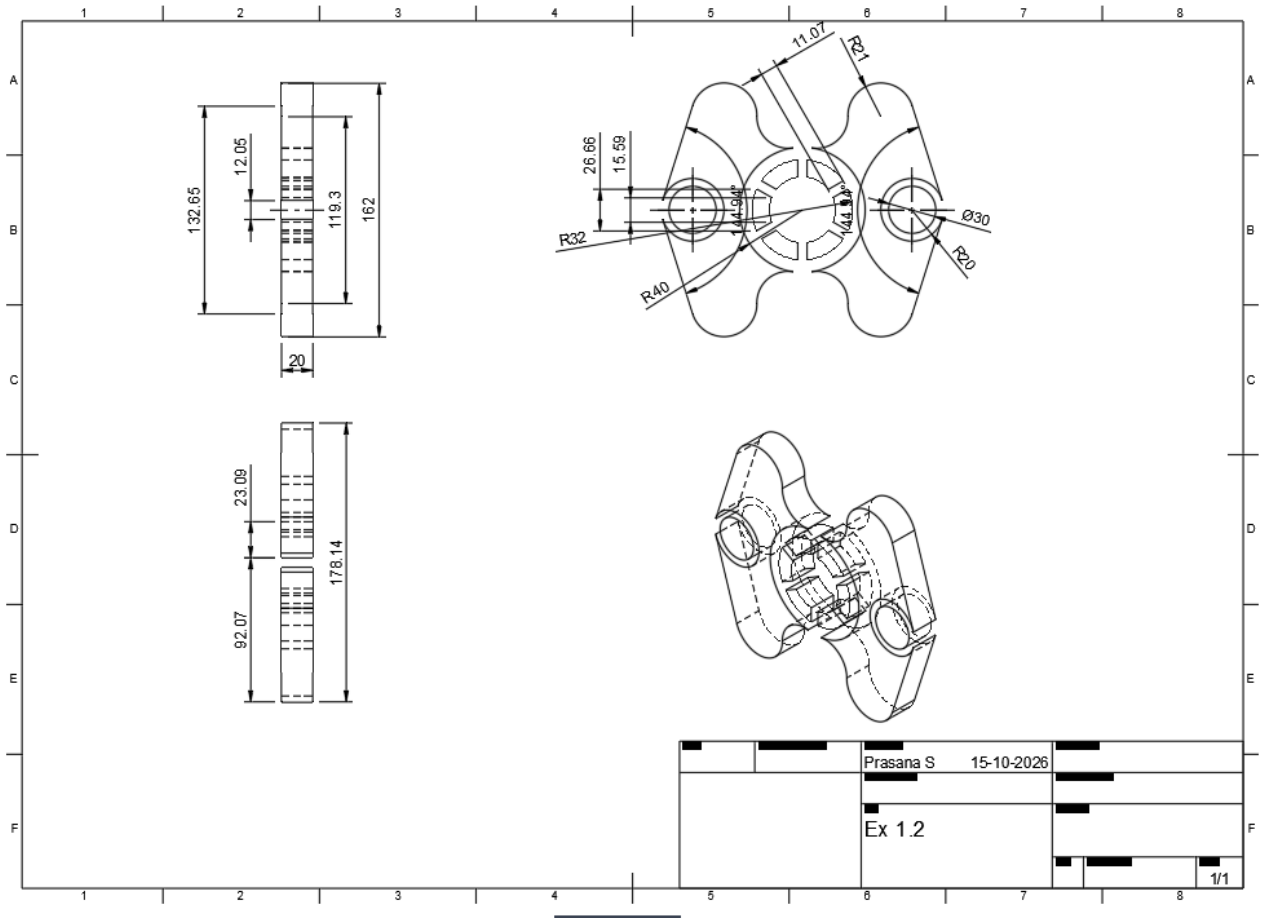


Figure 3: Dimensioned technical drawing

RESULT

Thus, the given component was sketched, dimensioned, modelled and drafted using Autodesk Fusion 360.